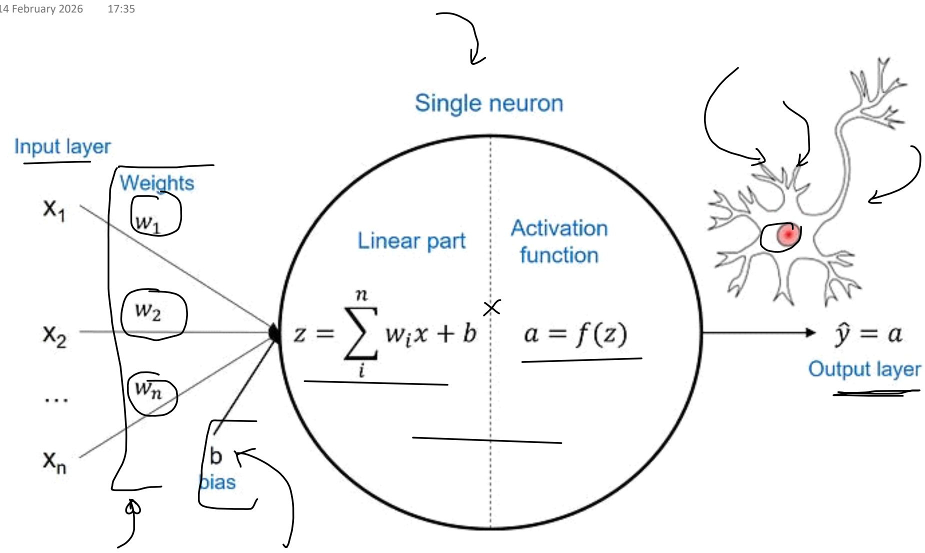
Neural Network (MLP-FFN)

14 February 2026 17:35



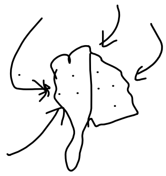
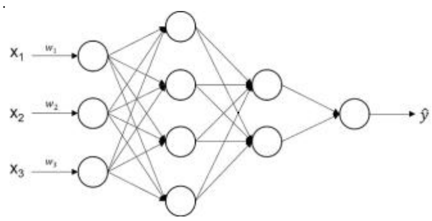
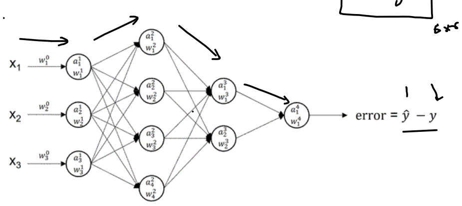


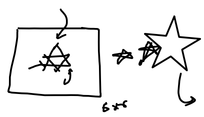
Image pixel
Text token
freq → $\frac{1}{7}$



- CNN
- RNN
- LSTM
- GRU
- Transformer →
- VIT

ffn





DL

Pytorch Tensorflow

Keras

Python

activation function

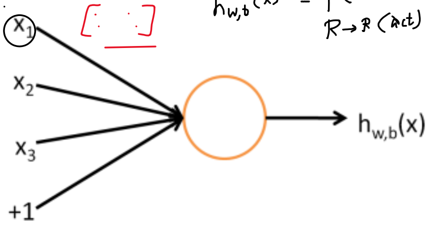
input $\rightarrow$ [ ] $\rightarrow$ o/p class

cost function

(x^i, y^i)

$h_{w,b}(x) = f(W^T x)$

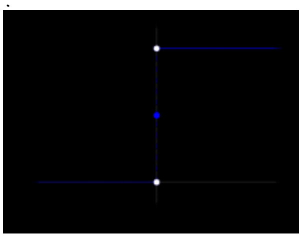
$R \rightarrow R$ (Act)



$y = \sum (W^x I) + b$

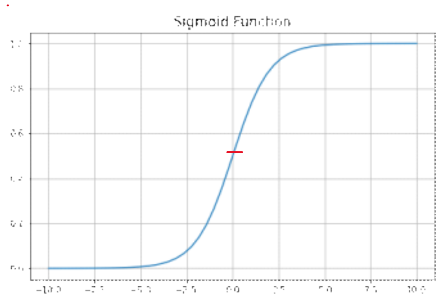
$(-\infty, +\infty)$

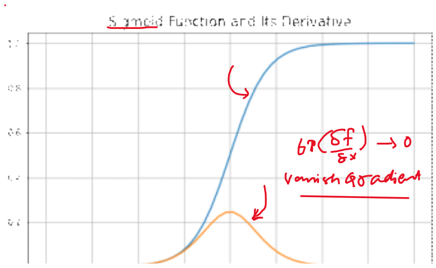
STEP $\quad y = cx$



1. Range
2. PD

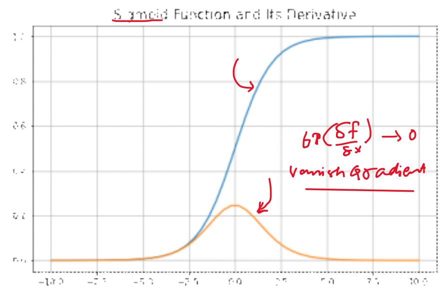
$\frac{1}{1+e^{-x}}$

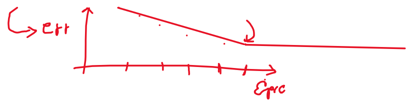




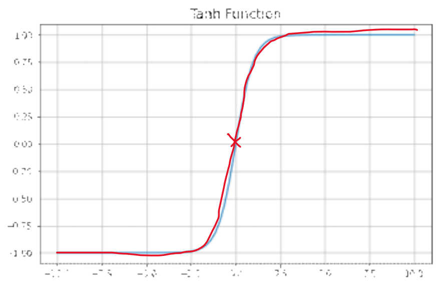
$67 \left(\frac{\delta f}{\delta x}\right) \rightarrow 0$

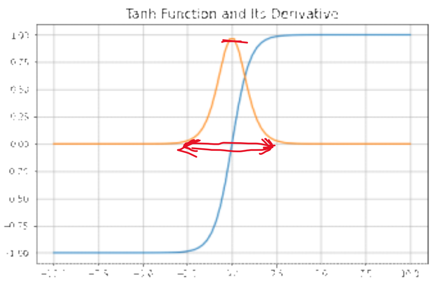
Vanish Gradient

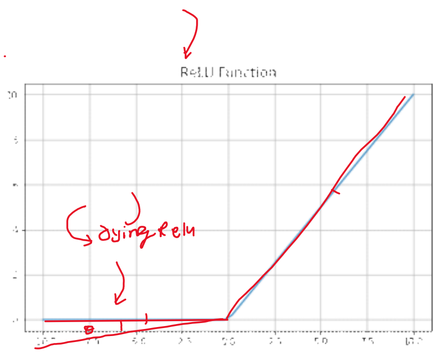


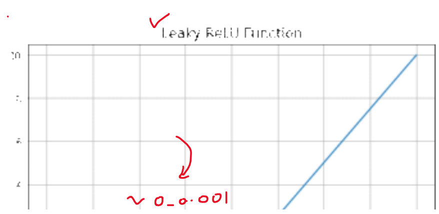


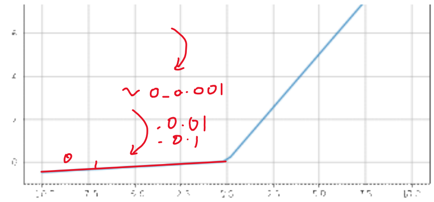
2. Tanh $\rightarrow (-1,1)$

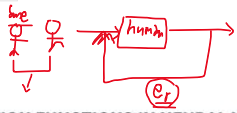


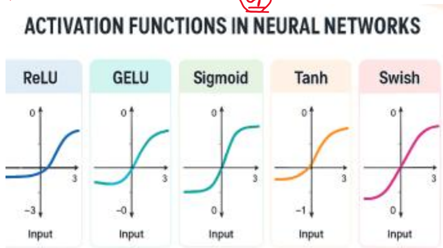


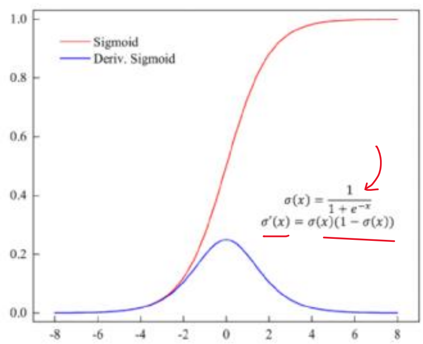






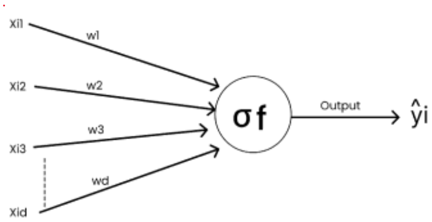






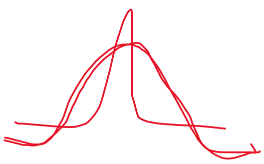
classification

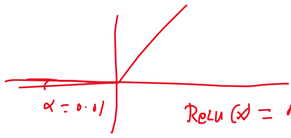
↳ $0 < \sigma(z) < 1$



$-1 < \tanh(h) < 1$

$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$





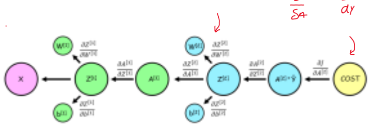
$\alpha = 0.01$

$ReLU(x) = \max(0, x)$ ↳ 0

$LeakyL(x) = \max(\alpha x, x)$ ↳ 0.01

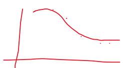
↳ $f(w, b) \to$ (Sigmoid) $\to$

$\frac{\delta}{\delta A}$ $\frac{dx}{dy}$

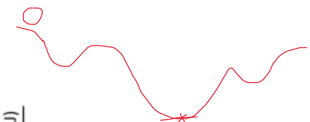


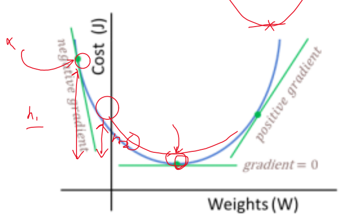
→ o/p

← Loss_cost function



EPOC
Batch



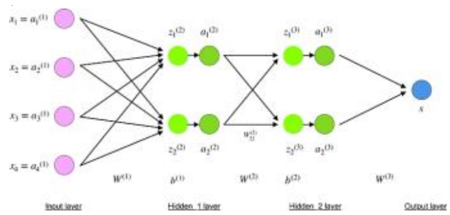


$W_{new} = W_{OLD} - \alpha \frac{dJ}{dW}$

$Cost(J) = \frac{1}{2}(\hat{Y} - Y)^2$

$$W^{[1]}_{new} = W^{[1]}_{old} - \alpha \frac{dJ}{dW^{[1]}} \quad | \quad b^{[1]}_{new} = b^{[1]}_{old} - \alpha \frac{dJ}{db^{[1]}}$$

$$W^{[2]}_{new} = W^{[2]}_{old} - \alpha \frac{dJ}{dW^{[2]}} \quad | \quad b^{[2]}_{new} = b^{[2]}_{old} - \alpha \frac{dJ}{db^{[2]}}$$



$x = a^{(1)} \quad Input\ layer$

$z^{(2)} = W^{(1)}x + b^{(1)} \quad neuron\ value\ at\ Hidden_1\ layer$

$a^{(2)} = f(z^{(2)}) \quad activation\ value\ at\ Hidden_1\ layer$

$z^{(3)} = W^{(2)}a^{(2)} + b^{(2)} \quad neuron\ value\ at\ Hidden_2\ layer$

$a^{(3)} = f(z^{(3)}) \quad activation\ value\ at\ Hidden_2\ layer$

$s = W^{(3)}a^{(3)} \quad Output\ layer$

$$\frac{\partial C}{\partial x} = [\frac{\partial C}{\partial x_1}, \frac{\partial C}{\partial x_2}, \ldots, \frac{\partial C}{\partial x_m}]$$

$$\frac{\partial C}{\partial w^l_{jk}} = \frac{\partial C}{\partial z^l_j}\frac{\partial z^l_j}{\partial w^l_{jk}} \qquad chain\ rule$$

$$z^l_j = \sum_{k=1}^{m} w^l_{jk}a^{l-1}_k + b^l_j \qquad by\ definition$$

$m \ - \ number\ of\ neurons\ in\ l-1\ layer$

$$\frac{\partial z^l_j}{\partial w^l_{jk}} = a^{l-1}_k \qquad by\ differentiation\ (calculating\ derivative)$$

$$\frac{\partial C}{\partial w^l_{jk}} = \frac{\partial C}{\partial z^l_j}a^{l-1}_k \qquad final\ value$$